VARUNA | OPERATIONAL SITUATION REPORT 01

Trishuli River flood, Rasuwa and Nuwakot, Nepal | Event: 26 August 2026 | Assessment issued: October 2026

Observation	Pre-event SAR	Post-event SAR	Domain
Sentinel-1 IW VV, orbit 19	24 Aug 2026, 00:18 UTC	05 Sep 2026, 00:18 UTC	1,526.05 km²

Situation overview. VARUNA compared two authentic Sentinel-1 GRD acquisitions processed with ESA SNAP and Copernicus 30 m elevation data. The post-event acquisition occurred 10 days after the flood; therefore the mapped surface anomalies represent persistent changes, not maximum inundation at flood peak. All exposure values below are spatial-screening estimates, not field-confirmed damage.

Table 1. Key assessment outputs (raw, unfiltered classification unless specified).

Candidate floodwater	Candidate debris / scour	Potential building exposure	Potential road exposure
126.60 km² (8.30%)	108.21 km² (7.09%)	1,062 / 3,816 (27.83%)	35.12 / 260.71 km (13.47%)

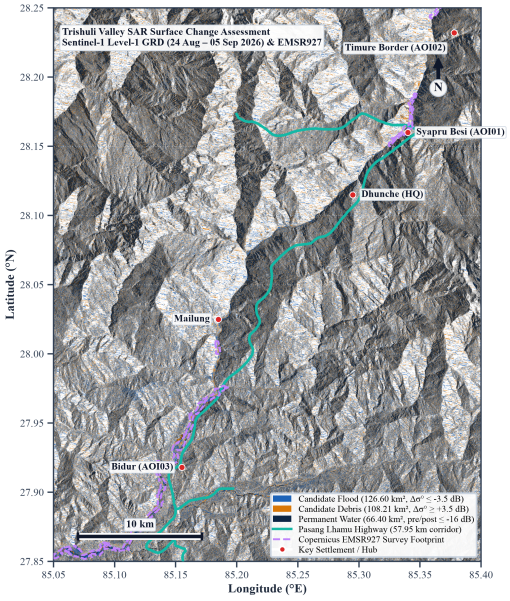


Figure 1. Sentinel-1 bitemporal change-screening map for the Trishuli corridor. High-resolution orthorectified geospatial export with candidate flood and debris screening overlay.

1. Independent Reference Check

Copernicus EMSR927 optical event polygons were used as held-out spatial reference data. SAR anomaly overlap was 37.16% in AOI03 (Bidur; 2.19 km² of a 5.89 km² optical footprint) and 23.70% in AOI01 (Syapru Besi; approximately 0.26 km² of a 1.11 km² footprint). These percentages measure time-offset spatial coincidence, not model accuracy or flood recall.

2. Infrastructure Screening

Potentially exposed assets include 1,062 buildings and 35.12 km of surveyed roads. The Pasang Lhamu Highway screening layer intersects 9.62 km of candidate hazard. Of the EMSR927 “Destroyed” building reference class, 936 of 2,813 mapped structures intersect radar-anomaly pixels. No road closure, structural collapse or settlement isolation should be treated as confirmed without field verification.

3. Operational Constraints and Recommended Use

Timing: September 5 imagery was collected 10 days after the event. Terrain: layover/shadow auxiliary bands were not preserved in the exported single-band radar raster. Calibration: SNAP issued a COG-wrapper LUT diagnostic disclosed in the research report. Use: prioritize candidate locations for inspection, compare with EMSR927 and local observations, and avoid making closure or evacuation decisions from SAR anomalies alone.

Data provenance. Copernicus Sentinel-1 GRD (CDSE); Copernicus 30 m DEM; Copernicus EMSR927 Rapid Mapping; OpenStreetMap and Nepal Survey Department vector layers. Methodology and complete numerical tables: VARUNA Research Report, October 2026. Contains modified Copernicus Sentinel data (2026). © OpenStreetMap contributors.